PART II: PATTERNS

Topic 8: Seasonality — Predictable Patterns

The Core Concept

Seasonality refers to patterns that repeat at **fixed, predictable intervals** — every week, every quarter, or every year. Unlike noise, seasonal patterns are reliable and anticipatable.

Why It Matters

If we do not account for seasonality, we will misread routine fluctuations as crises — and waste resources reacting to predictable patterns we could have planned for.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Year	Period	Precinct North (UoF)	Precinct South (UoF)	North Arrests
2021	Jan–Mar	28, 26, 30	25, 24, 26	~500 avg
	Apr–Jun	32, 35, 40	24, 25, 23	
	Jul–Sep	42, 38, 34	26, 25, 24	
	Oct–Dec	30, 28, 25	25, 26, 24	
2022	Jan–Mar	22, 20, 24	24, 26, 25	~450 avg
	Apr–Jun	26, 28, 32	23, 25, 24	
	Jul–Sep	35, 30, 28	25, 26, 24	
	Oct–Dec	24, 22, 18	24, 25, 23	
2023	Jan–Mar	12, 10, 14	25, 22, 28	~410 avg
	Apr–Jun	15, 18, 22	24, 26, 24	
	Jul	25	27	700
	Aug	**(Missing)**	23	680
	Sep	19	22	500
	Oct	16	**85 ★ Outlier**	420
	Nov–Dec	14, 11	24, 26	~370 avg

Look at Precinct North's July peak across three years:

- **2021:** 42 incidents (July peak)
- **2022:** 35 incidents (July peak)
- **2023:** 25 incidents (July peak)

The **timing** is identical every year — that is seasonality. The **level** is declining year over year — that is trend. Both patterns coexist.

Likely drivers: heat, longer daylight hours, more people outdoors, more police-public contact opportunities.